

Exercise 05

Relational algebra and functional dependencies

Given the following relations:

Students:	SName	SMatrikel	SBirthday
	Heintje	2143	05.04.2001
	Eva	3333	06.04.2001
	Luise	3334	06.04.2002
	Daniel	3335	07.04.2002
	Daniel	3336	07.04.2001
	Heintje	3337	17.04.2001

Lecturers:	LName	LOffice	LTel
	Klaus	C201	123
	Maria	D22	null
	Marlene	C204	456
	Matze	E4	null

Event:	ENAME	ESemester	ERoom	ELecturer
	Beachvolleyball	ss17	Beach	Maria
	Beachvolleyball	ss18	Beach	Maria
	Hanggliding	ss17	Beach	Maria
	Hanggliding	ss18	Beach	Maria
	Sackhüpfen	ws17	null	Klaus
	Sackhüpfen	ws18	null	Klaus
	Dance Gymnastics	ss17	D111	Klaus
	Dance Gymnastics	ss18	D111	Klaus

Task 1: Anomalies

Specify a deletion anomaly, insertion anomaly and update anomaly for the relation `Lecturers`.

Even such small examples can already lead to problems with redundancies. Saving offices in the same relation as `Lecturers` and not moving offices into a separate relation results in a deletion anomaly, for example when the tuple with lecturer Maria is deleted. The information that the office D22 even exists is lost in this case. In addition, the relation `Lecturers` is not suitable for storing new offices without lecturers. In that case, the lecturer would always be `null` and we want the lecturers to be the primary key. Thus, despite the `Office` attribute, we cannot add new (empty) offices (insertion anomaly). If we remodel our buildings and the office-identifiers change, for example now C201 is suddenly called C2.2.1, then we generate an update anomaly if we only change the identifier for Klaus' office, but not for Marlene's.

Task 2: Keys, superkeys and foreign keys

Justify the following statements and show them formally by applying the definitions:

- a) `SName` in `Students` cannot be a key.
Daniel appears twice, but has a different matriculation number. Formally, every subset of $X \subseteq \{SName\} \subseteq Students$ must differ in at least one attribute value. But $t_4(SName) = t_5(SName) = \text{Daniel}$.
- b) `LName` from `Lecturers` does not meet the foreign key constraint for `ELecturer` in `Events`.
It's exactly the other way round. `ELecturer` points to `LName`. Formally: $\{t(LName) \mid t \in Lecturers\} \subseteq \{t(ELecturer) \mid t \in Events\}$ must apply. However, this is not fulfilled for $t_4(LName) = \text{Matze}$.
- c) `ELecturer` in `Events` fulfils the foreign key constraint for `LName` from `Lecturers`.
This way is correct because $\{t(ELecturer) \mid t \in Events\} \subseteq \{t(LName) \mid t \in Lecturers\}$
- d) `ESemester`, `ERoom`, `ELecturer` in `Events` fulfils not the superkey property in the given instance of the database.
Maria has hang-gliding and beach volleyball on the beach in the summer semester 2018.
Formally:
 $\{ESemester, ERoom, ELecturer\} \subseteq Events$ always differ in at least one attribute value. But $t_2(ESemester, ERoom, ELecturer) = t_4(ESemester, ERoom, ELecturer)$
- e) `LName`, `LOffice` fulfils the superkey property of `Lecturers` in the given instance of the database.
 $\{LName, LOffice\}$ fulfils the super key property because $\{LName, LOffice\} \subseteq Lecturers$ always differ in at least one attribute value.
- f) `LOffice` in `Lecturers` fulfils the key constraint.
The attribute values of `LOffice` have no double entries and therefore it fulfils the key constraint. We cannot tell from the data that `LOffice` is a key, that is domain knowledge.
- g) The relational schema `R` is a superkey of `R`.
Every relational schema $R = A_1, \dots, A_n$ fulfils the superkey property of `R`, since $A_1, \dots, A_n$ always differs in at least one attribute value (set semantics of relational algebra)
- h) `ENAME`, `ESemester` fulfils the superkey constraint of `Events`.
 $\{ENAME, ESemester\}$ fulfils the superkey attribute of `Events` because there are no two tuples with $t_i(ENAME, ESemester) = t_j(ENAME, ESemester)$ in `Events`.

Task 3*: Functional dependencies, Splitting Cover and Key Heuristic

Starting from our example using Lecturers, Students, Events and Grades we now assume that there are only the following attributes: $SName$, $SMatr$, $SBDay$, $LName$, $LOffice$, $LTel$, $EName$, $ESemester$, $ERoom$ and $Grade$ stored in one relation R (yes, that's bad!).

We assume, that the following functional dependencies FD are valid:

$$\begin{aligned} SMatr &\rightarrow SMatr, SName, SBDay \\ LName &\rightarrow LOffice, LTel \\ EName, ESemester &\rightarrow ERoom \\ SMatr, EName, ESemester &\rightarrow Grade \\ SMatr &\rightarrow SName \\ LName, LOffice &\rightarrow LTel \\ SMatr, EName, ESemester &\rightarrow ERoom \end{aligned}$$

- a) Simplify this set FD using the SPLITTING algorithm.

$$\begin{aligned} SMatr &\rightarrow SName, SBDay \\ LName &\rightarrow LOffice, LTel \\ EName, ESemester &\rightarrow ERoom \\ SMatr, EName, ESemester &\rightarrow Grade \\ LName, LOffice &\rightarrow LTel \\ SMatr, EName, ESemester &\rightarrow ERoom \end{aligned}$$

- b) Provide 10 FD s that are in $\{FD\}^+$ for the functional dependencies FD that are not in the list above.

$$\begin{aligned} \text{Triviality} &: SName \rightarrow SName \\ &SName, LName \rightarrow SName, LName \\ \text{Reflexivity} &: SName, LName \rightarrow SName \\ \text{Augmentation} &: LName, LOffice, SMatr \rightarrow LTel \\ &SMatr, LTel, LOffice \rightarrow SName \\ &SMatr, LTel, LOffice \rightarrow SName, LOffice \\ \text{Decomposition} &: LName \rightarrow LTel \\ &SMatr \rightarrow SName \end{aligned}$$

- c) What is $\{SMatr\}^{+_{FD}}$?
 $\{SMatr\}^+_F = \{SMatr, SName, SBDay\}$
- d) What is $\{SMatr, LName\}^{+_{FD}}$?
 $\{SMatr, LName\}^+_F = \{SMatr, LName, SName, SBDay, LOffice, LTel\}$
- e) What is $\{SName, LOffice, ERoom\}^{+_{FD}}$?
 $\{SName, LOffice, ERoom\}^+_F = \{SName, LOffice, ERoom\}$

- f) Simplify the set from part (a) of the task further using the COVER algorithm. What can you see, when you compare the FDs to the relations of Task 1?

SPLITTING: done.

Minimise left sides, we do not need to check single-element sets

$ENAME, ESEMESTER \rightarrow EROOM$

$ENAME \in \{ESEMESTER\}^+_{FC}$? No.

$ESEMESTER \in \{ENAME\}^+_{FC}$? No.

$LNAME, LOFFICE \rightarrow LTEL$

$LTEL \in \{LNAME\}^+_{FC}$? Yes, simplify to $LNAME \rightarrow LTEL$

$SMATR, ENAME, ESEMESTER \rightarrow EROOM$

$EROOM \in \{ENAME, ESEMESTER\}^+_{FC}$? Yes, simplify to $ENAME, ESEMESTER \rightarrow EROOM$

Remove unnecessary functional dependencies

none

Result:

$SMATR \rightarrow SNAME, SBDAY$

$LNAME \rightarrow LOFFICE, LTEL$

$ENAME, ESEMESTER \rightarrow EROOM$

$SMATR, ENAME, ESEMESTER \rightarrow GRADE$

Every FD belongs to one of the Relation in Task 1 $\rightarrow$ BCNF!

- g) Determine the keys using Key Heuristic.

SPLITTING: done.

Attributes that do not appear on any right side are in the key: SMATR, LNAME, ENAME, ESEMESTER

Test whether keys are found: $\{MATRICULATION, LNAME, ENAME, ESEMESTER\}^+_F = \{MATRICULATION, LNAME, ENAME, ESEMESTER, SNAME, SDATE_of_birth, LOFFICE, LTEL, EROOM, GRADE\} = RDATA$

Key found, $\{MATRICULATION, LNAME, ENAME, ESEMESTER\}$ is the only key of RDATA